

Qatar-10b

R-0128

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-10_250704 · created 2026-07-16T05:59:41+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460860.695 +/- 0.01 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.046
Transit depth (Rp/Rs)^2	3.0 +/- 1.59 [%]
Orbital Inclination (inc)	89.16 +/- 2.49
Airmass coefficient 1 (a1)	0.741 +/- 0.04
Airmass coefficient 2 (a2)	0.23 +/- 0.12
Scatter in the residuals of the lightcurve fit is	5.46 %
Best Comparison Star	#1 - [521, 324]
Optimal Aperture	2.27
Optimal Annulus	5.08
Transit Duration (day)	0.084 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.173 ± 0.046 vs 0.1265 (deviation 0.65σ)
Duration fit vs published	0.084 d vs 0.1154 d (deviation 1.83σ)
Mid-time O–C	13.1 min
Depth SNR	2.4
Residual scatter	5.46 %

Light curve

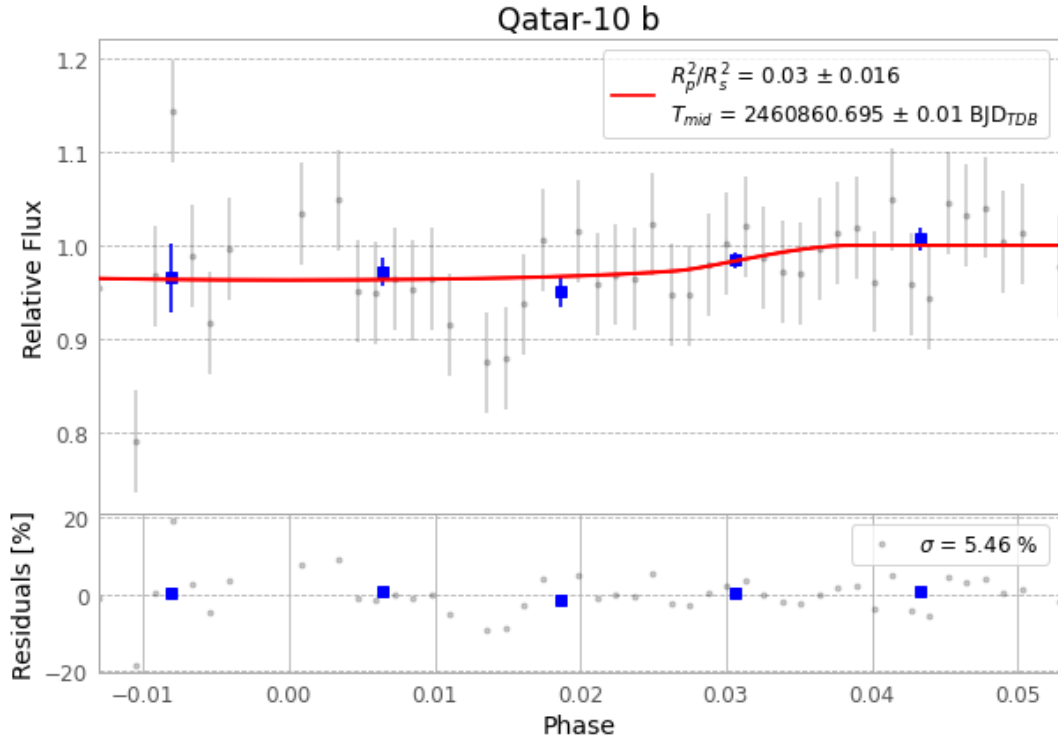


Figure 1 — FinalLightCurve_Qatar-10 b_2025-07-03.png (SHA-256 14bef2f349bc30e2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [412, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9c63ea84b6ce21b1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

53 FITS frames (35 MB), Qatar-10250704040915.FITS → Qatar-10250704064515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-10-250704.log (790c853ffa21...), FinalLightCurve_Qatar-10 b_2025-07-03.png (14bef2f349bc...), FinalLightCurve_Qatar-10 b_2025-07-03.csv (147f9c2c0994...)

Compute resources

Wall time 140.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 05:59Z.